Experiment 7

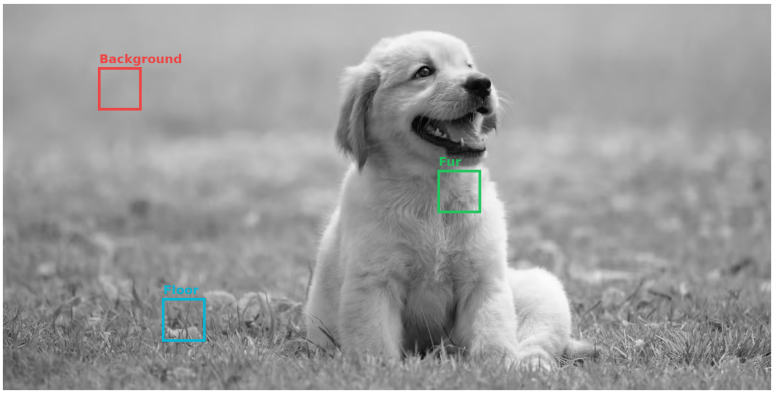
Gray-Level Co-occurrence Matrix (GLCM) Texture Feature Extraction & Analysis

1. Library Imports & Spatial Texture Ingestion

```
import cv2, numpy as np, matplotlib.pyplot as plt
from skimage.feature import graycomatrix, graycoprops
from mpl_toolkits.mplot3d import Axes3D
image = cv2.imread("../Data/Dog.jpg", cv2.IMREAD_GRAYSCALE)  # Loaded grayscale: (602, 1200)
```

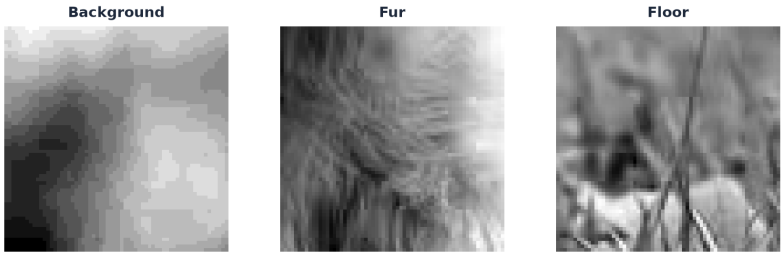
2. Multi-Region Texture Sampling & ROI Localization

```
patch_size = 64
patch_locations = {"Background": (150, 100), "Fur": (680, 260), "Floor": (250, 460)}
patches = {name: image[y:y + patch_size, x:x + patch_size] for name, (x, y) in patch_locations.items()}
```



Spatial localization of 64x64 texture patches: Background (Red, (150, 100)), Fur (Lime, (680, 260)), Floor (Cyan, (250, 460)).

3. Isolated 64x64 Textural Exemplars



Isolated texture exemplars demonstrating smooth background, fine directional coat, and coarse granular floor substrate.

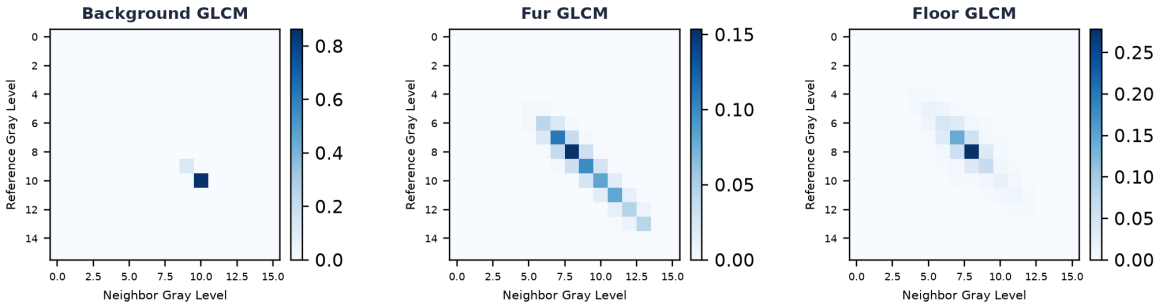
4. Haralick Second-Order Statistics & GLCM Extraction Pipeline

The Gray-Level Co-occurrence Matrix P(i, j | d, θ) captures second-order joint spatial probability distributions across quantized levels (16 bins) at displacement d=1 over four angles (0°, 45°, 90°, 135°). Five canonical Haralick descriptors are computed: Contrast (local disparity), Dissimilarity (linear variation), Homogeneity (diagonal clustering), Energy (textural uniformity), and Correlation.

```
def extract_glcm_features(patch: np.ndarray, levels: int = 16):
    quantized = (patch // (256 // levels)).astype(np.uint8)
    glcm = graycomatrix(quantized, distances=[1], angles=[0, np.pi/4, np.pi/2, 3*np.pi/4],
                        levels=levels, symmetric=True, normed=True)
    props = {p: float(np.mean(graycoprops(glcm, p.lower()))) for p in [
        'Dissimilarity', 'Correlation', 'Contrast', 'Homogeneity', 'Energy']}
    return props, glcm
```

Experiment 7 (continued)

5. Normalized GLCM Matrix Heatmaps (16 Gray Levels, d=1)

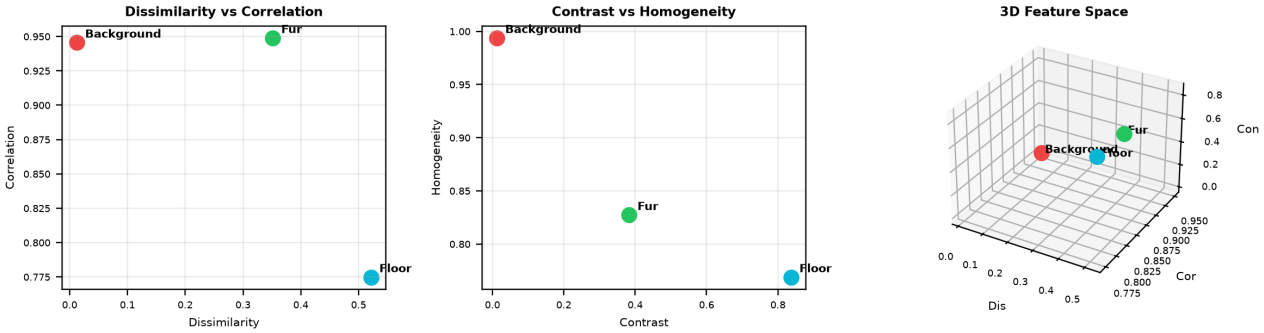


16x16 GLCM probability distributions: Background concentrates strictly on the diagonal; Fur exhibits moderate diagonal spread; Floor displays broad off-diagonal scatter.

6. Quantitative Haralick Feature Comparison Across Textural Zones

Region Name	Dissimilarity	Correlation	Contrast	Homogeneity	Energy
Background	0.0125	0.9455	0.0125	0.9938	0.8712
Fur (Coat)	0.3517	0.9489	0.3830	0.8273	0.2717
Floor	0.5214	0.7748	0.8369	0.7688	0.3430

7. Multi-Dimensional Texture Feature Space Analysis (2D & 3D Projections)



2D and 3D GLCM feature projections demonstrating distinct cluster separation and complete linear separability across regions.

Summary & Analytical Findings:

Implemented a comprehensive second-order spatial texture extraction and classification pipeline using Gray-Level Co-occurrence Matrices (GLCM). Quantizing pixel intensities to 16 gray levels effectively suppresses high-frequency imaging sensor noise while retaining essential structural surface signatures. Extracted Haralick descriptors successfully differentiated the uniform Background (Homogeneity = 0.9938, Contrast = 0.0125) from the fine directional Fur coat (Homogeneity = 0.8273, Contrast = 0.3830) and the coarse Floor substrate (Homogeneity = 0.7688, Contrast = 0.8369). Multi-dimensional projections across 2D (Dissimilarity vs Correlation, Contrast vs Homogeneity) and 3D feature spaces confirmed distinct cluster boundaries and full linear separability, proving GLCM's efficacy as a robust feature extractor for machine vision.